CS 340 README Template

About the Project
Grazioso Salvare is an innovative international rescue-animal training company that identifies dogs suitable for search-and-rescue training. This software application can work with existing data from animal shelters to identify and categorize available dogs for training. The application includes a database.

Motivation
Grazioso Salvare needs to identify dogs with specific profiles suitable for different types of rescue missions, such as water rescue, mountain rescue, disaster recovery, and scent tracking. The project aims to improve this process by using data from local animal shelters.

Getting Started
Installation prerequisites:

- **MongoDB**: We need MongoDB to store and manage our data in a document-oriented database. You can download it from the official MongoDB website.
- **Python 3+**: Programming language for writing and executing our data processing scripts. You can download it from the official Python website.
- **Jupyter Notebook**: Use Jupyter Notebook to write and run our Python code, visualize data, and much more. You can install it using pip (after installing Python) by following the steps in the official Jupyter website.

Follow the Installation steps to setup the project.

Installation

1. Clone the repository or download the files. This repository includes the following files:

- animal_shelter.py
- ProjectTwoDashboard.ipynb
- Grazioso Salvare Logo.png

2. Install **pymongo** to connect our Python code to the MongoDB database and perform CRUD database operations from our scripts:

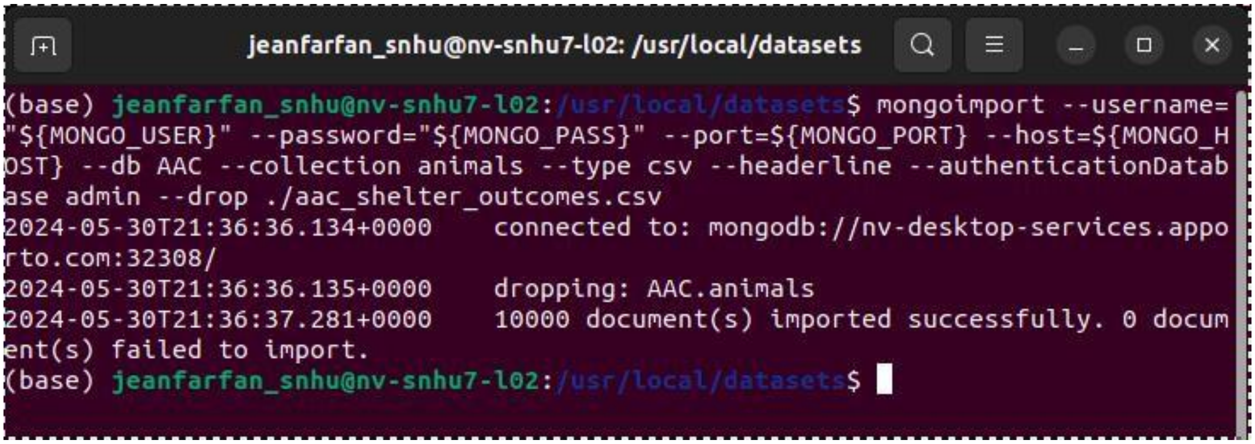
   pip install pymongo

3. Import the ***aac_shelter_outcomes.csv*** file into MongoDB:

   mongoimport –username=”${MONGO_USER}” --password=”${MONGO_PASS}” --port=${MONGO_PORT} --host=${MONGO_HOST} --db AAC --collection animals --type csv --headerline –authenticationDatabase admin --file path/to/aac_shelter_outcomes.csv

   (Note: If the command throws an error, make sure that the .csv file directory is correct and that you are using the correct MongoDB user).

Note: This template has been adapted from the following sample templates: Make a README, Best README Template, and A Beginners Guide to Writing a Kickass README.



4. Edit the connection variables in ***animal_shelter.py*** file to match your MongoDB:

   USER = 'your_user'
   PASS = 'your_password'
   HOST = 'your_host'
   PORT = your_port
   DB = 'AAC'
   COL = 'animals'

Create a Mongo user with read and write privileges for our database (optional)

1. Open the terminal and enter the MongoDB shell with the following command:

mongosh

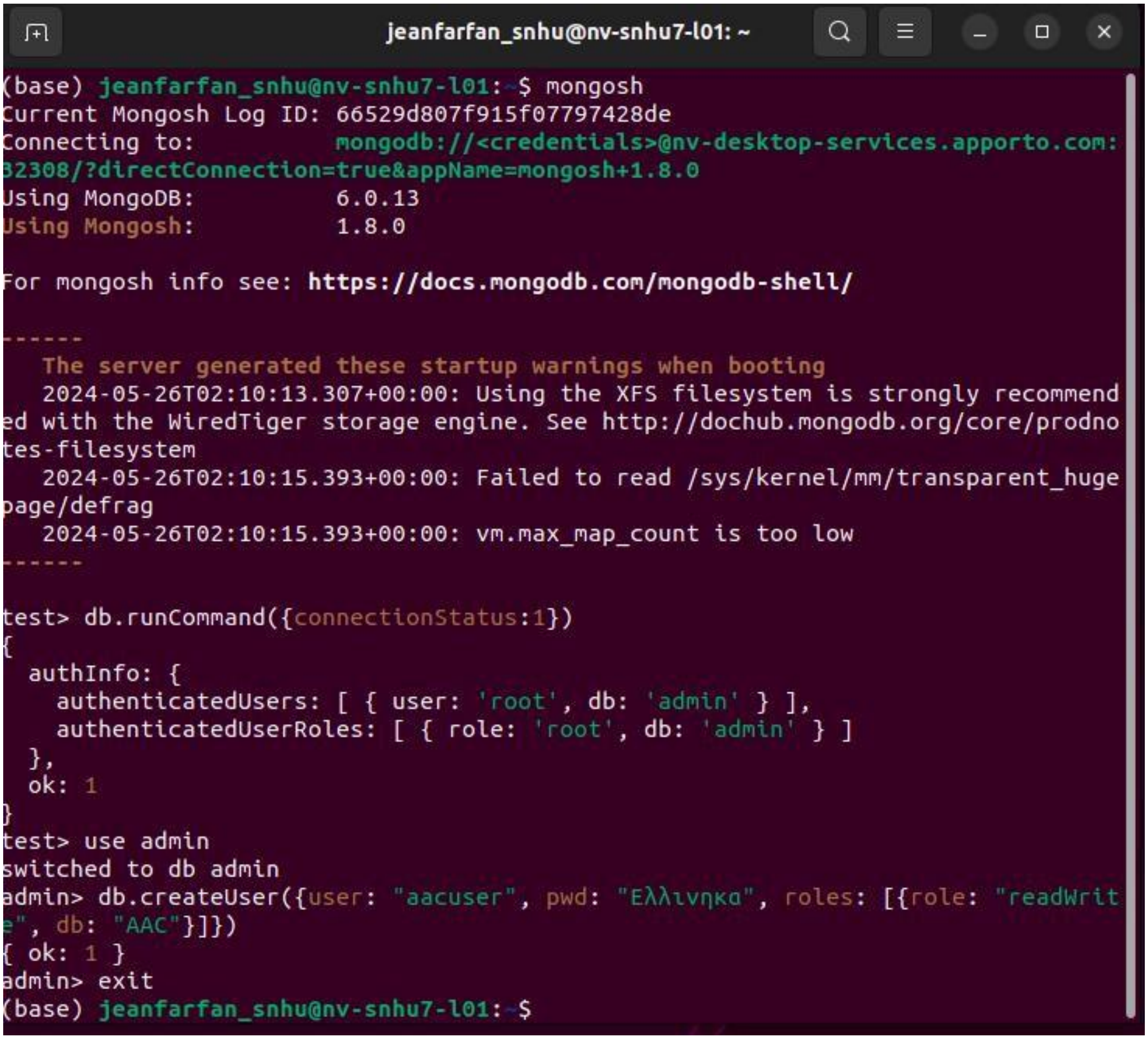
2. Use admin database:

use admin

3. Create user:

db.createUser({user: “your_username”, pwd: “your_password”, roles: [{role: “readWrite”, db: “AAC”}]})

Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().



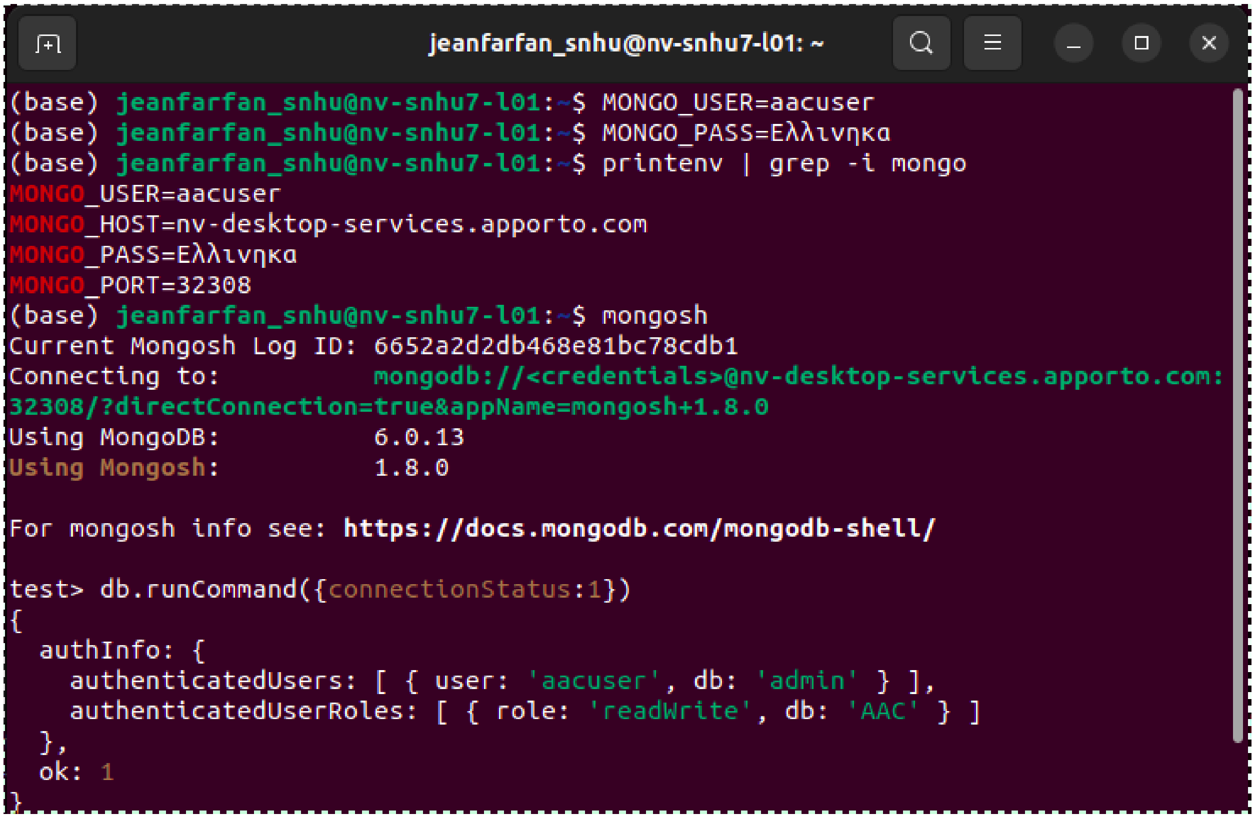
4. Exit MongoDB shell and change Mongo user in the terminal:

MONGO_USER=your_username
MONGO_PASS=your_password

5. Try to open the MongoDB shell and execute the following command for verification:

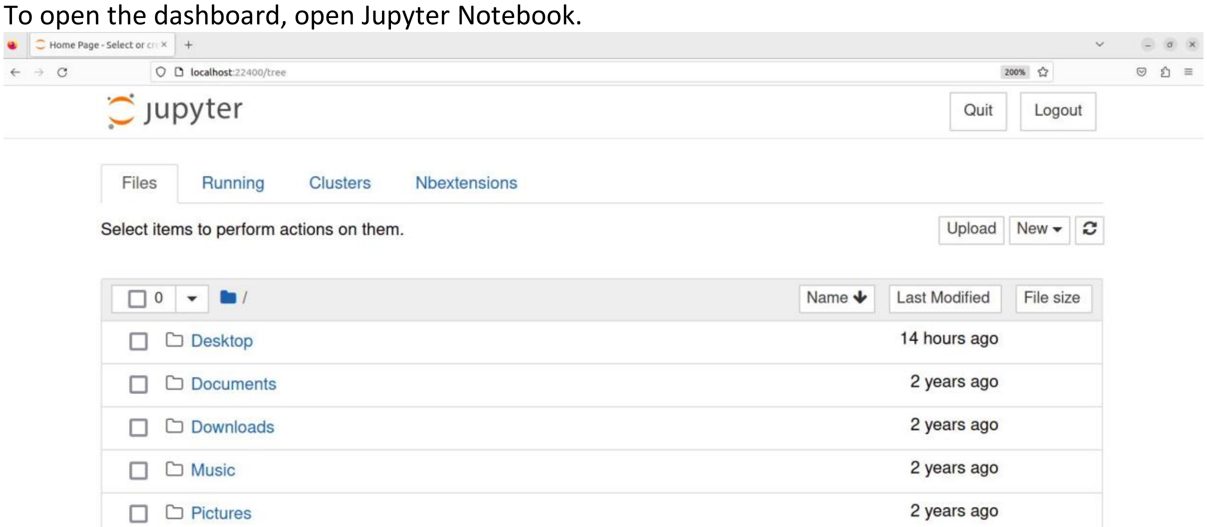
db.runCommand({connectionStatus:1})

Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().



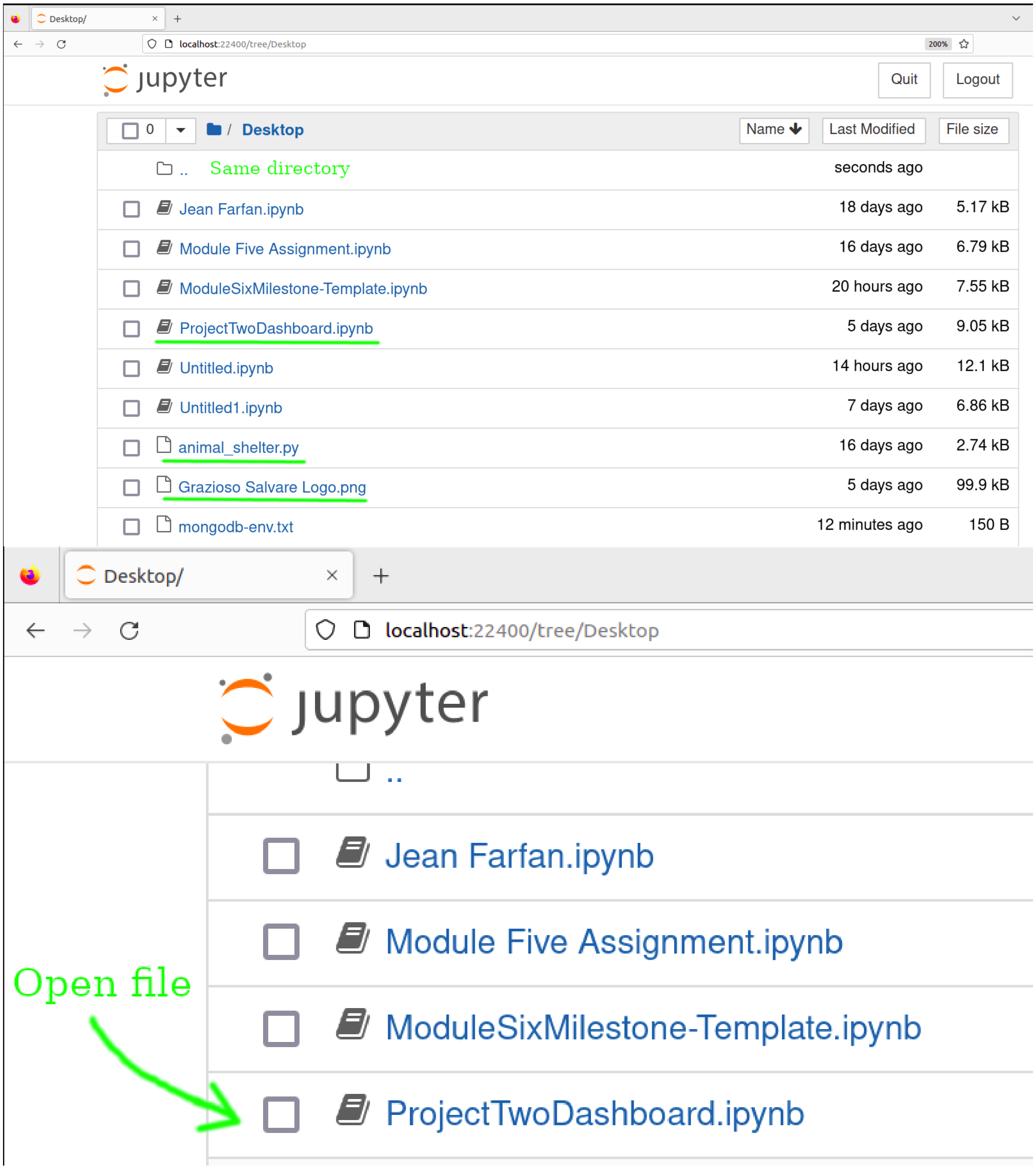
Open dashboard

1. To open the dashboard, open Jupyter Notebook.



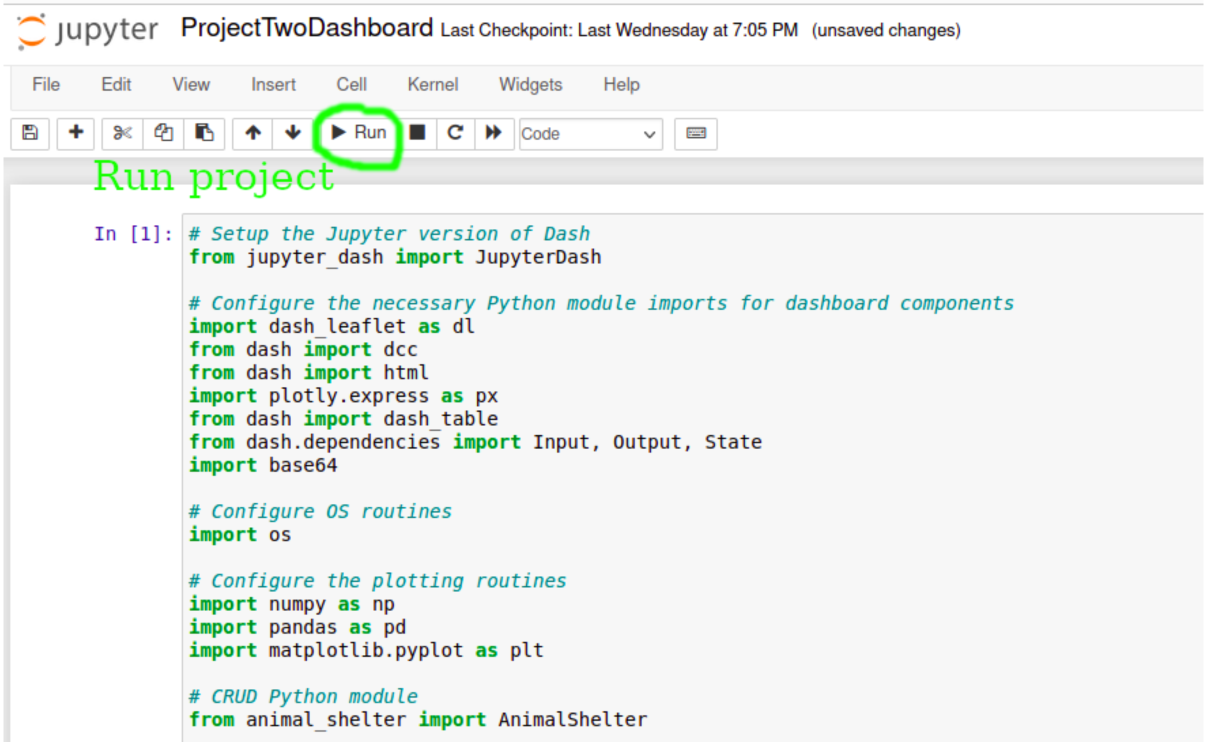
2. Search and open the ***ProjectTwoDashboard.ipynb*** file. Make sure that all the files (***ProjectTwoDashboard.ipynb***, ***Grazioso Salvare Logo.png***, and ***animal_shelter.py***) are in the same directory.

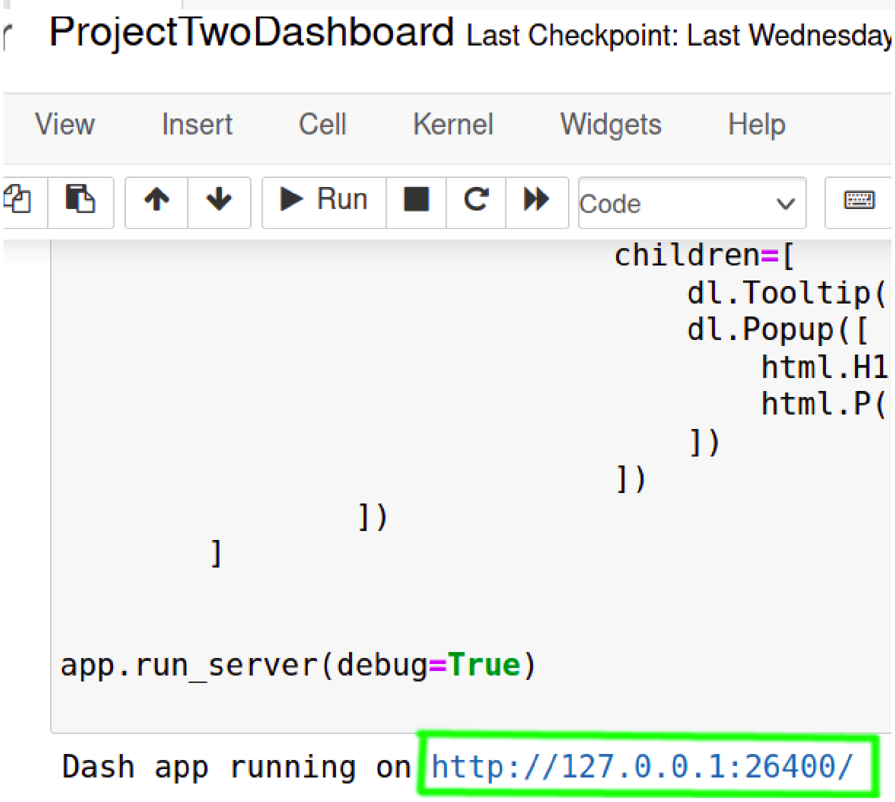
Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().



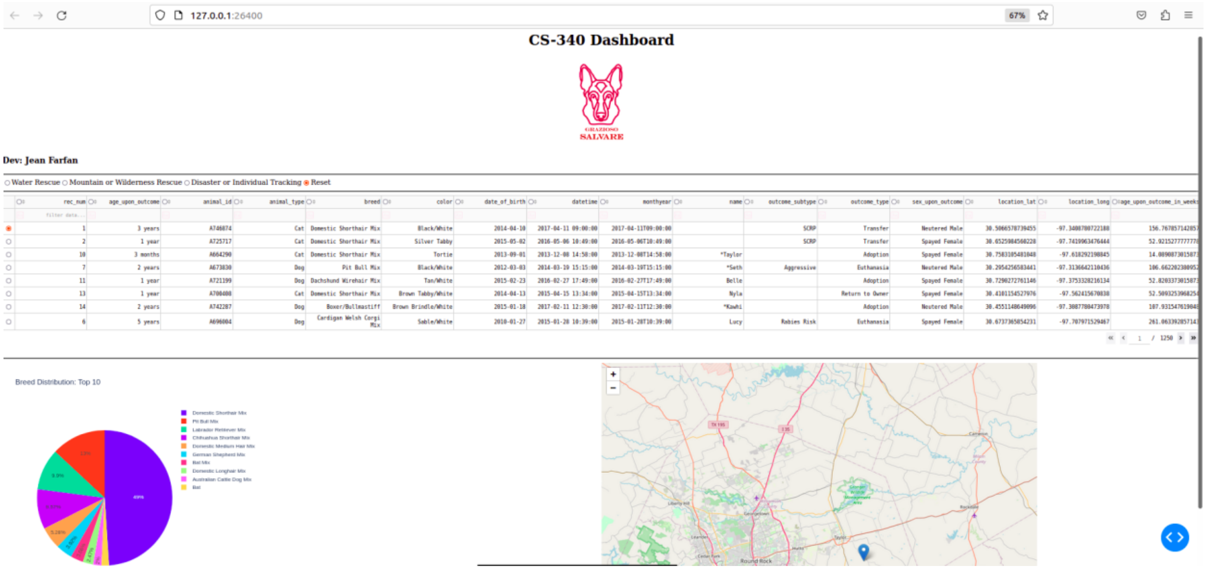
3. Run the file to show the dashboard. This will launch the dashboard on a local server at http://127.0.0.1:8050/.

Note: This template has been adapted from the following sample templates: Make a README, Best README Template, and A Beginners Guide to Writing a Kickass README.





Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().



Usage

The dashboard developed for Grazioso Salvare uses data from local animal shelters to help identify dogs suitable for various rescue missions. This dashboard provides an interactive interface to filter and visualize the data. This dashboard allows users to filter dogs based on the type of rescue mission. Also, it displays the filtered data in an interactive data table and a pie chart for better analysis and decision-making. Finally, it supports CRUD (Create, Read, Update, and Delete) operations on the dog data stored in MongoDB.

Use interactive radio items to filter the data based on the type of rescue mission. The data table and charts will update dynamically to reflect the filtered data.

To perform Create, Read, Update, and Delete operations on the data, you need to write code in ***ProjectTwoDashboard.ipynb*** or in a new Python file. Ensure the MongoDB connection variables in animal_shelter.py are correctly configured to match your MongoDB setup. Below is an example of how to use CRUD methods using the provided ***animal_shelter.py***.

Code Example

```
from animal_shelter import AnimalShelter

shelter = AnimalShelter()  # Create an instance of AnimalShelter

# Example animal document
animal = {
  "age_upon_outcome": ‘99 years’,
  "animal_id": 'B666777',
  "animal_type": 'Cuy',
  "breed": 'Domestic Shorthair Mix',
  "color": 'Orange',
```

Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().

```
 "date_of_birth": '1925-06-13',
 "datetime": '2018-04-11 09:00:00',
 "monthyear": '2019-04-11T09:00:00',
 "name": 'Gamo',
 "outcome_subtype": 'SCRP',
 "outcome_type": 'Transfer',
 "sex_upon_outcome": 'Neutered Male',
 "location_lat": 30.5066578739454,
 "location_long": -97.3408780722187,
 "age_upon_outcome_in_weeks": 156.767857142856
}

# Create
# Create a new animal
print(shelter.create(animal))

# Find
# Read an animal
results = shelter.read({"animal_id": "B666777"})
for k in results:
   print(k)

# Update
# First, key/value to find
criteria = {"animal_id" : "B666777"}
# Then, key/value to modify
updateData = {"age_upon_outcome": '77 years'}
# print modified count
print(shelter.update(criteria, updateData))
# Find the updated animals to see results
results = shelter.read({"animal_id" : "B666777"})
# Print all animals with animal_id B666777
for k in results:
   print (k)
# Delete
# Delete animal with animal_id B666777
print(shelter.delete({"animal_id" : "B666777"}))
# Find deleted animals to see results
results = shelter.read({"animal_id" : "B666777"})
# Print all animals with animal_id B666777
for k in results:
   print (k)
```

Tests

To validate the functionality, launch the dashboard and use the interactive widgets to filter data, and perform CRUD operations via the provided interface. This will ensure that the application meets the requirements for identifying and categorizing dogs suitable for various rescue missions.

Note: This template has been adapted from the following sample templates: Make a README, Best README Template, and A Beginners Guide to Writing a Kickass README.

Screenshots

```
In [1]: from animal_shelter import AnimalShelter
```

```
In [2]: shelter = AnimalShelter() # class object
        animal = { # document
          "age_upon_outcome": '99 years',
          "animal_id": 'B666777',
          "animal_type": 'Cuy',
          "breed": 'Domestic Shorthair Mix',
          "color": 'Orange',
          "date_of_birth": '1925-06-13',
          "datetime": '2018-04-11 09:00:00',
          "monthyear": '2019-04-11T09:00:00',
          "name": 'Gamo',
          "outcome_subtype": 'SCRP',
          "outcome_type": 'Transfer',
          "sex_upon_outcome": 'Neutered Male',
          "location_lat": 30.5066578739454,
          "location_long": -97.3408780722187,
          "age_upon_outcome_in_weeks": 156.767857142856
        }
```

```
In [3]: print(shelter.create(animal))

        True
```

```
In [4]: results = shelter.read({"animal_id" : "B666777"})
        # Print all animals with animal_id B666777
        for k in results:
            print (k)

        {'_id': ObjectId('6659568f6e1cd69e37204665'), 'age_upon_outcome': '99 years', 'animal_id': 'B666777', 'animal_type':
        'Cuy', 'breed': 'Domestic Shorthair Mix', 'color': 'Orange', 'date_of_birth': '1925-06-13', 'datetime': '2018-04-11
        09:00:00', 'monthyear': '2019-04-11T09:00:00', 'name': 'Gamo', 'outcome_subtype': 'SCRP', 'outcome_type': 'Transfer
        ', 'sex_upon_outcome': 'Neutered Male', 'location_lat': 30.5066578739454, 'location_long': -97.3408780722187, 'age_u
        pon_outcome_in_weeks': 156.767857142856}
```

Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().

```python
In [13]: # First, key/value to find
criteria = {"animal_id" : "B666777"}
# Then, key/value to modify
updateData = {"age_upon_outcome": '77 years'}
```

```python
In [14]: print(shelter.update(criteria, updateData))
```

```
1
```

```python
In [15]: # Find the updated animals to see results
results = shelter.read({"animal_id" : "B666777"})
# Print all animals with animal_id B666777
for k in results:
    print (k)
```

```
{'_id': ObjectId('66625b88ccd5db01393ede82'), 'age_upon_outcome': '77 years', 'animal_id': 'B666777', 'animal_type': 'Cuy', 'breed': 'Domestic Shorthair Mix', 'color': 'Orange', 'date_of_birth': '1925-06-13', 'datetime': '2018-04-11 09:00:00', 'monthyear': '2019-04-11T09:00:00', 'name': 'Gamo', 'outcome_subtype': 'SCRP', 'outcome_type': 'Transfer ', 'sex_upon_outcome': 'Neutered Male', 'location_lat': 30.5066578739454, 'location_long': -97.3408780722187, 'age_upon_outcome_in_weeks': 156.767857142856}
```

```python
In [16]: # Delete animal with animal_id B666777
print(shelter.delete({"animal_id" : "B666777"}))
```

```
1
```

```python
In [17]: # Find deleted animals to see results
results = shelter.read({"animal_id" : "B666777"})
# Print all animals with animal_id B666777
for k in results:
    print (k)
```

```python
In [ ]:
```

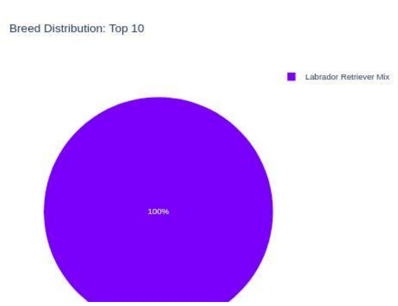
127.0.0.1:26400

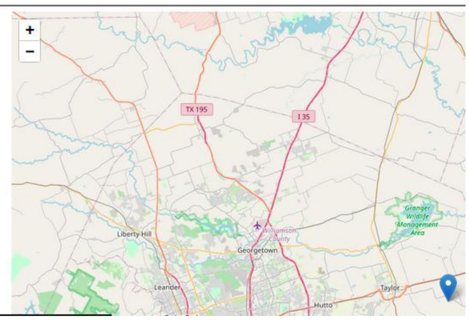
GRAZIOSO
SALVARE

Dev: Jean Farfan

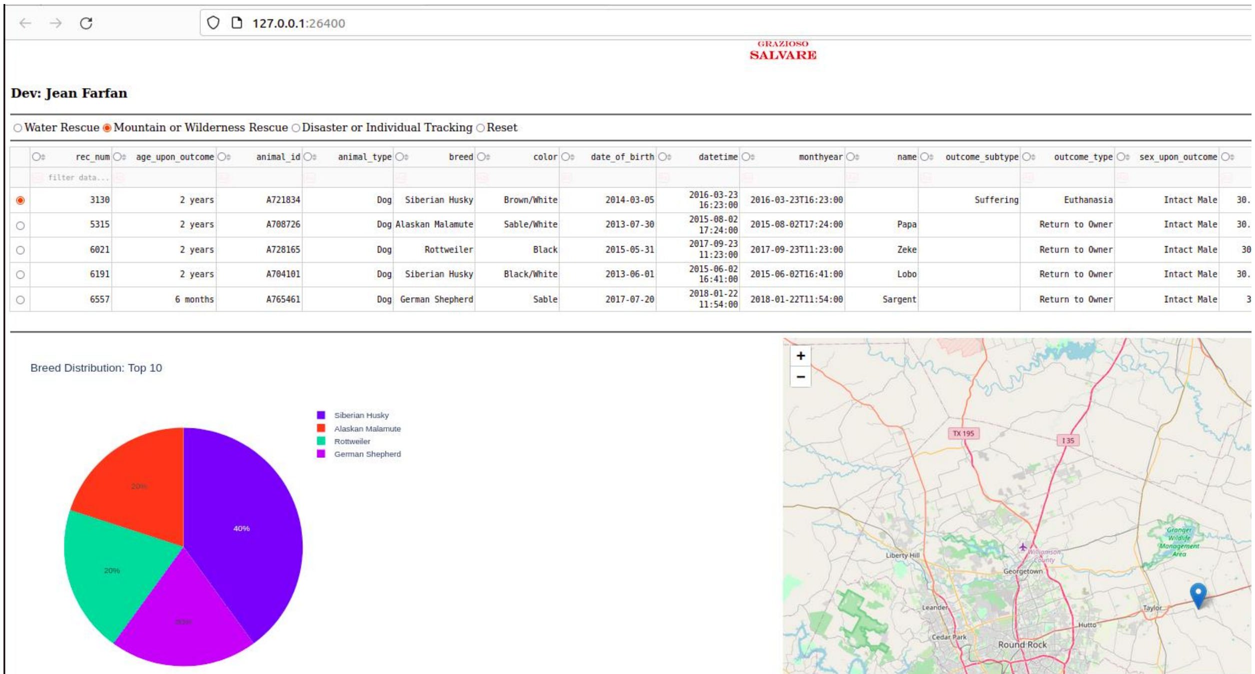
Water Rescue | Mountain or Wilderness Rescue | Disaster or Individual Tracking | Reset

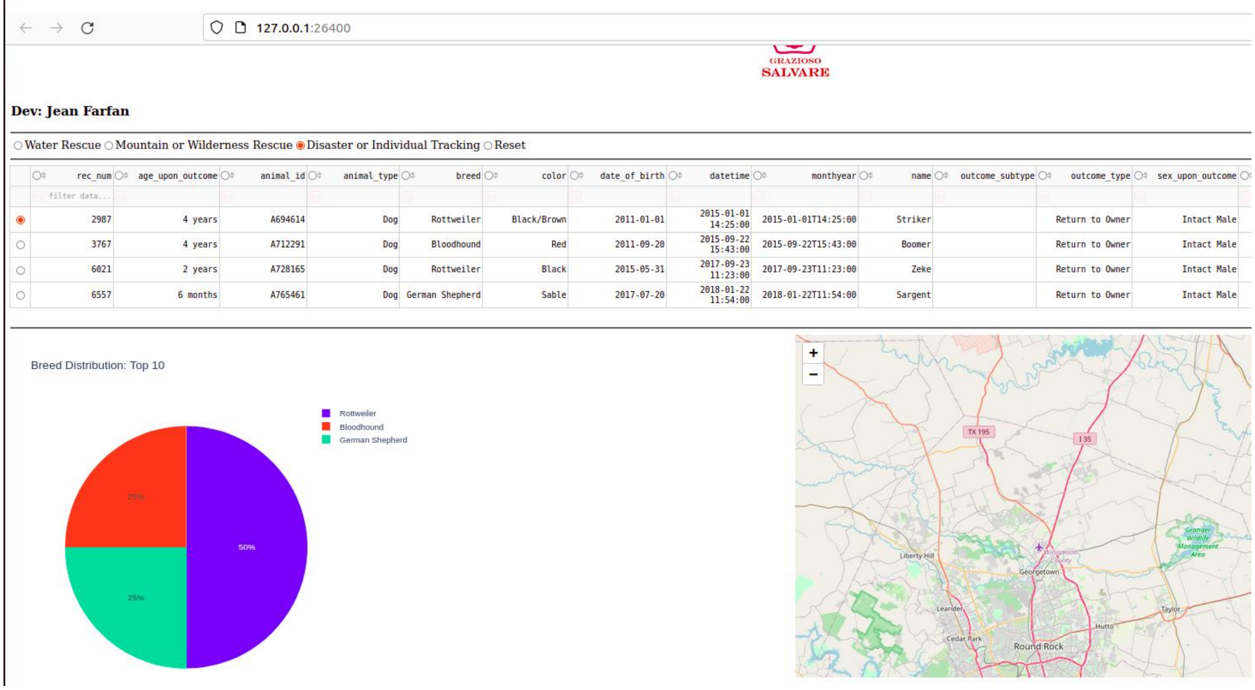
	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome
	filter data...												
●	36	6 months	A706953	Dog	Labrador Retriever Mix	Yellow	2014-12-06	2015-07-06 11:33:00	2015-07-06T11:33:00		Medical	Euthanasia	Intact Female
○	732	2 years	A749782	Dog	Labrador Retriever Mix	Tan/White	2015-05-19	2017-07-25 14:59:00	2017-07-25T14:59:00	*Catalina		Return to Owner	Intact Female
○	1121	1 year	A757158	Dog	Labrador Retriever Mix	White/Black	2016-08-30	2017-08-31 14:12:00	2017-08-31T14:12:00	Pirata		Return to Owner	Intact Female
○	1628	9 months	A740471	Dog	Labrador Retriever Mix	Tan/White	2016-03-17	2016-12-23 17:13:00	2016-12-23T17:13:00	Mika		Adoption	Intact Female
○	1757	7 months	A742767	Dog	Labrador Retriever Mix	Black	2016-06-27	2017-02-14 15:20:00	2017-02-14T15:20:00	Marley		Return to Owner	Intact Female
○	1988	1 year	A762781	Dog	Labrador Retriever Mix	Black/White	2016-11-27	2017-12-03 13:09:00	2017-12-03T13:09:00		Partner	Transfer	Intact Female
○	2041	2 years	A702745	Dog	Labrador Retriever Mix	Black	2013-05-22	2015-05-22 11:45:00	2015-05-22T11:45:00	Abigail		Return to Owner	Intact Female
○	2225	2 years	A757341	Dog	Labrador Retriever Mix	Black/White	2015-09-01	2017-10-03 12:27:00	2017-10-03T12:27:00	19	Partner	Transfer	Intact Female



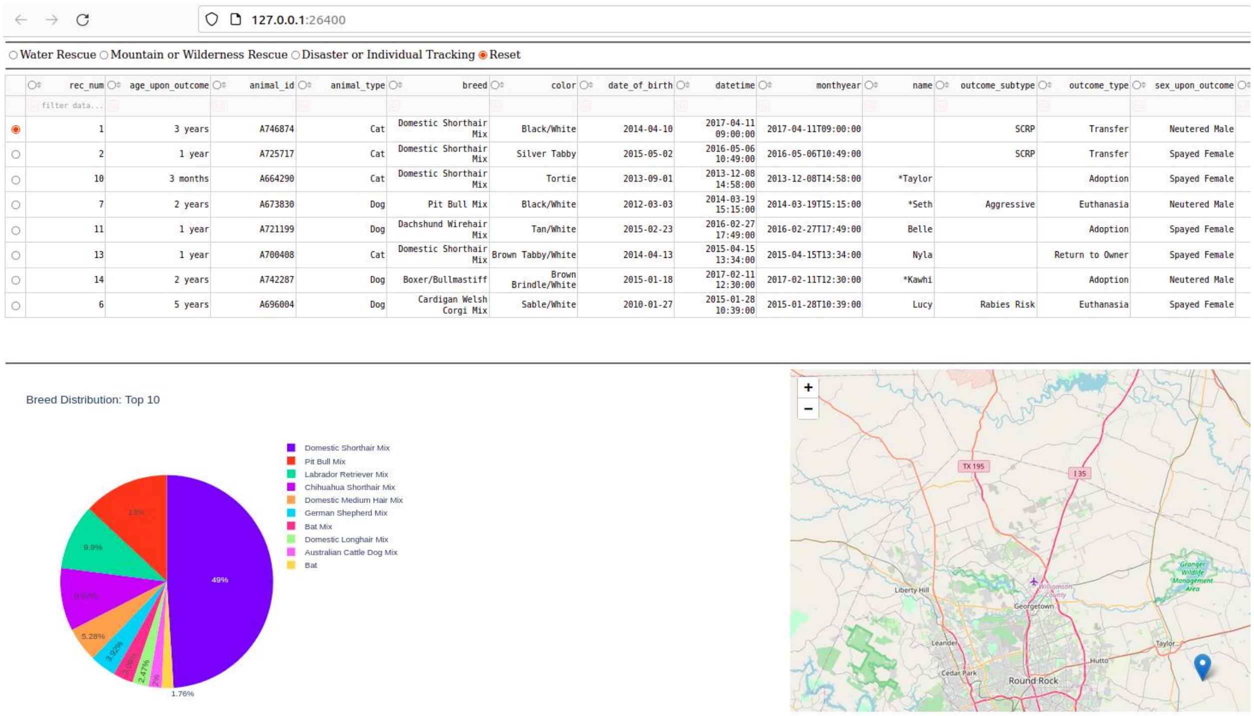


Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().





Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().



127.0.0.1:26400

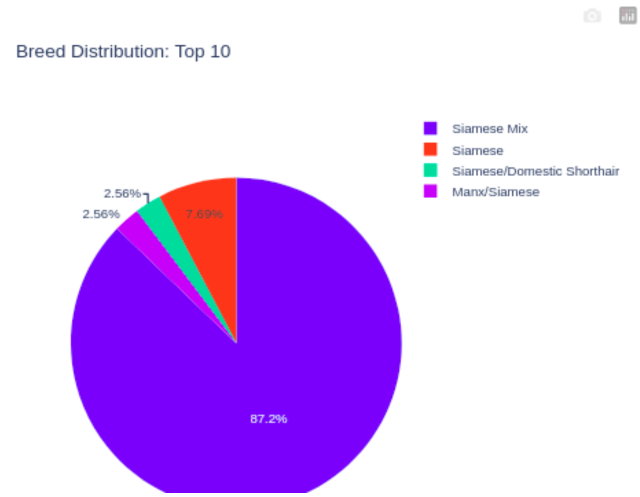
Water Rescue | Mountain or Wilderness Rescue | Disaster or Individual Tracking | Reset

rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome
1	3 years	A746874	Cat	Domestic Shorthair Mix	Black/White	2014-04-10	2017-04-11 09:00:00	2017-04-11T09:00:00		SCRP	Transfer	Neutered Male
2	1 year	A725717	Cat	Domestic Shorthair Mix	Silver Tabby	2015-05-02	2016-05-06 10:49:00	2016-05-06T10:49:00		SCRP	Transfer	Spayed Female
10	3 months	A664290	Cat	Domestic Shorthair Mix	Tortie	2013-09-01	2013-12-08 14:58:00	2013-12-08T14:58:00	*Taylor		Adoption	Spayed Female
7	2 years	A673830	Dog	Pit Bull Mix	Black/White	2012-03-03	2014-03-19 15:15:00	2014-03-19T15:15:00	*Seth	Aggressive	Euthanasia	Neutered Male
11	1 year	A721199	Dog	Dachshund Wirehair Mix	Tan/White	2015-02-23	2016-02-27 17:49:00	2016-02-27T17:49:00	Belle		Adoption	Spayed Female
13	1 year	A700408	Cat	Domestic Shorthair Mix	Brown Tabby/White	2014-04-13	2015-04-15 13:34:00	2015-04-15T13:34:00	Nyla		Return to Owner	Spayed Female
14	2 years	A742287	Dog	Boxer/Bullmastiff	Brown Brindle/White	2015-01-18	2017-02-11 12:30:00	2017-02-11T12:30:00	*Kawhi		Adoption	Neutered Male
6	5 years	A696004	Dog	Cardigan Welsh Corgi Mix	Sable/White	2010-01-27	2015-01-28 10:39:00	2015-01-28T10:39:00	Lucy	Rabies Risk	Euthanasia	Spayed Female

Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().

○ Water Rescue ○ Mountain or Wilderness Rescue ○ Disaster or Individual Tracking ◉ Reset

	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth
	filter data...		A6	Cat	Siamese		
◉	616	5 months	A684362	Cat	Siamese Mix	Tortie Point	2014-05-26
○	843	1 month	A680409	Cat	Siamese Mix	Flame Point	2014-05-02
○	1074	1 year	A667027	Cat	Siamese Mix	Flame Point	2012-11-11
○	1350	2 months	A683412	Cat	Siamese Mix	Flame Point	2014-05-10
○	1472	3 years	A665943	Cat	Siamese Mix	Lynx Point	2010-10-24
○	1492	17 years	A698600	Cat	Siamese Mix	Lynx Point	1998-03-14
○	1623	2 years	A696517	Cat	Siamese Mix	Lynx Point	2013-02-06
○	1706	10 years	A672606	Cat	Siamese Mix	Lynx Point	2004-02-16



Contact

Your name: Jean Farfan

Note: This template has been adapted from the following sample templates: [Make a README](), [Best README Template](), and [A Beginners Guide to Writing a Kickass README]().